3 **(a)** State the principle of conservation of momentum.

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..... [1]

(b) State the relationship between impulse and the force-time graph for a body.

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..... [1]

(c) A ball of mass 0.20 kg was initially moving to the right on a smooth floor with a velocity of 5.0 m s^{-1} before a rightward force *F* was applied to the ball. Fig. 3.1 shows the variation with time of the force *F* applied to an object.

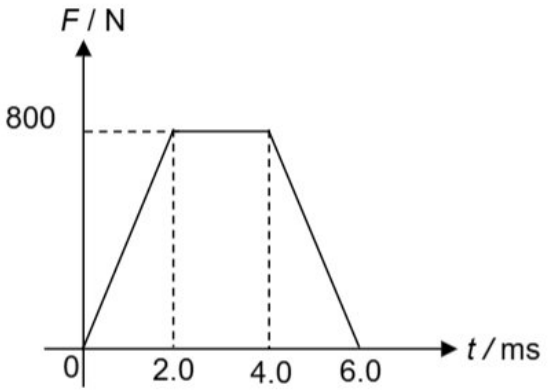


Fig. 3.1

(i) Determine the final momentum of the ball.

momentum = ... N s [3]

(ii) Taking rightwards as positive, sketch the variation with time of the momentum p of the ball on Fig. 3.2.

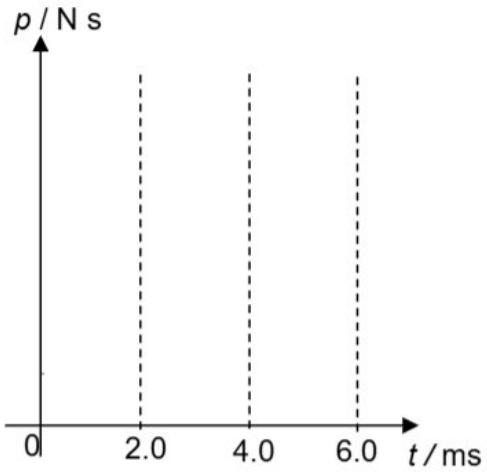


Fig. 3.2

[2]

(d) The ball from part **(c)** now collides with a wall.

With reference to the principle of conservation of momentum, state and explain whether momentum of the **ball** is conserved during collision.

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..... [2]